1 Privileged motion-mimic teacher

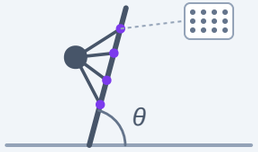
RL with full simulator state

Motion reference
IK-generated 4-beat cycle,
paced dwells ≥ 0.3 s
phase command c_t (in o_t)

RSI

PPO
r: motion mimic + climb
+ microspine safety
+ regularizers

Wall-climb sim



staged θ curriculum: 45° → 90°

microspine contact model
3×4 spine units per pad

domain randomization

o_t
x_t
a_t

teacher policy π_T
actor–critic MLP 256–128–64
21-D joint targets @ 50 Hz
trained

privileged x_t (sim only)
per-pad spine engagement (4)
per-pad seating fraction (4)
base height (1)

2 Estimator distillation

regress privileged state from proprio history (DAgger)

On-policy rollouts (DAgger)
frozen teacher π_T acts on (o_t , $\hat{e}_t$)
frozen

o_t
x_t
L = MSE($\hat{e}_t$, x_t)
$\hat{e}_t$
$\hat{e}_t$ into rollouts

proprio history H_t
50 frames × 85-D (1 s)

estimator φ
MLP 512–256
trained

estimator input is pure proprioception —
contact / load-cell channels ablated

3 RL fine-tuning on ê

PPO on π with the frozen estimator in the loop

Wall-climb sim
full DR
90° wall
mimic reward

PPO · 600 it
same mimic reward

o_t

policy π
init from π_T
critic keeps true x_t
trained

H_t

estimator φ
frozen

$\hat{e}_t$

a_t

Deployment
onboard (φ, π) + phase-clock table — proprioception only
ê-triggered clock-freeze safety on suspected detachment

proprioception (o_t, history H_t) · privileged state x_t (sim only) · estimate $\hat{e}_t = \varphi(H_t)$ · gradient update · reference / reward · trained · frozen